

Target-Free Action-Conditioned Predictive Diagnostics for Visually Stressed World Models

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Abstract

Visual world models can fail under visual corruption, yet a deployment-time diagnostic may have neither the true future nor a reliable planner cost. We introduce Action-Conditioned Predictive Consistency (ACPC), a post-hoc diagnostic that propagates nominal and perturbed histories under the same action sequence. The resulting rollout radius upper-bounds, sample by sample, the change in prediction error to any action-matched future. On four control tasks, correct-action eight-step features reduce held-out absolute future-drift mean absolute error by 56.5% relative to one-step features (95% cluster-bootstrap interval [51.1, 61.2]%) and by 51.9% relative to the strongest action/time-destroyed control ([45.0, 56.6]%). A region calibrated on LeWM seed 3072 reaches 0.945 balanced accuracy on two held-out training seeds; without stressor-specific refitting, a paired rule reaches 0.889 balanced accuracy and 0.864 Spearman correlation across blur and resize. ACPC therefore supports target-free mechanism analysis and family-local checkpoint calibration, but not universal thresholds, absolute robustness certification, or closed-loop guarantees.

Keywords: visual world models; predictive stability; action-conditioned diagnostics; calibration transfer; visual corruption.

Code and data release: <https://github.com/Anguo-star/lewm-acpc-diagnostics>

1 Introduction

Latent-predictive world models avoid reconstructing every pixel, but this does not by itself make their predictions robust. For control, nuisance variation is action dependent: a texture may be irrelevant, whereas a contact pose, doorway state, or object-goal relation must remain distinguishable if it changes the useful action. Encoder invariance is therefore only an upstream signal. The operational question is whether a same-state visual perturbation remains small after the encoder shift is propagated through the action-conditioned predictor.

This is a diagnostic study of frozen checkpoints, not a new robust-training method. ACPC pairs a nominal history with a visually perturbed view of the same state, applies one shared action sequence, and measures projected-rollout disagreement. Its value depends on three questions. *Mechanism*: does the disagreement control an action-matched future-facing error? *Calibration*: do thresholds fitted on one training seed transfer within a model family? *Stressor transfer*: does the calibrated score retain useful ordering beyond the calibration corruption? We evaluate these questions separately so checkpoint classification accuracy cannot stand in for action or horizon specificity.

We instantiate ACPC on LeWM [13] under Gaussian observation noise across TwoRoom, PushT, Reacher, and Cube. No-noise checkpoints remain strong on clean observations yet lose up to 74.56 ± 5.55 success points at evaluation noise $\sigma = 0.08$. Full-sequence Gaussian augmentation recovers much of this performance and provides a controlled family of fragile and recovered checkpoints. This family lets us test future-drift mechanism, held-out-seed calibration, and blur/resize transfer with distinct evidence units.

The contributions are:

1. **A target-free predictive bound with a matched empirical test.** ACPC upper-bounds samplewise error change to any action-matched future. On all four tasks and both primary test seeds, the correct-action $H = 8$ response improves future-drift prediction beyond $H = 1$ and structure-destroyed $H = 8$ controls.
2. **Within-family cross-seed calibration.** A joint radius/guard region fitted only on LeWM seed 3072 reaches balanced accuracy/AUPRC 0.945/0.965 on the same four-task grid from independently trained seeds 3073/3074, with no threshold refit.

3. **Transfer beyond the calibration corruption.** One paired improvement rule transfers without stressor-specific refitting from Gaussian calibration to fixed blur and resize, reaching balanced accuracy/AUPRC 0.889/0.974 and Spearman correlation 0.864.

2 Related Work

Latent-predictive world models. JEPA [11], I-JEPA [1], and V-JEPA [3, 2] learn by predicting latent targets rather than reconstructing pixels. LeWM [13], our model family, stabilizes an end-to-end JEPA world model with SIGReg [5]. Theory explains why latent prediction can suppress nuisance or noisy features [18, 12, 9], but does not make a learned checkpoint robust to every action-relevant visual shift.

Visual robustness and control-relevant representations. Data augmentation methods such as DrQ/DrQ-v2 [10, 22] and SODA [8] improve visual control by changing training. ACPC is instead a post-hoc diagnostic for a frozen checkpoint. Its selective-consistency motivation is related to bisimulation and value-aware models, which preserve distinctions with different downstream control consequences [6, 23, 7, 19, 17]. Our narrower intervention asks whether two views of the same state remain close after one shared action sequence while a proxy guard rejects gross representational collapse.

Action and dynamics consistency diagnostics. Recent work uses inverse probes, latent action decoding, cycle consistency, or generated-future compatibility to test action semantics [4, 24, 16, 14, 20]. MWM incorporates action-conditioned consistency during training [21], whereas iKCE diagnoses kinematic long-horizon failures [15]. ACPC instead intervenes on paired visual views of the same state and connects rollout disagreement to action-matched future-error drift; we do not claim priority for action consistency in general.

3 Target-Free Predictive Diagnostic

3.1 Action-matched rollout radius

Let h be a nominal observation-action history and $\tilde{h}$ a visually probed view of the same underlying state. A frozen encoder E_θ maps the histories to $z = E_\theta(h)$ and $\tilde{z} = E_\theta(\tilde{h})$. Both branches are propagated by the predictor F_θ under the *same* action sequence $\mathbf{a} = (a_0, \dots, a_{H-1})$. For a task-relevant projection Π and nonnegative weights $\sum_{k=1}^H \alpha_k = 1$, define

$$\bar{G}_{\mathbf{a}}(z) = [\sqrt{\alpha_1}\Pi(F_\theta^1(z, a_0)); \dots; \sqrt{\alpha_H}\Pi(F_\theta^H(z, \mathbf{a}))], \quad \text{ACPC}_H(h, \tilde{h}, \mathbf{a}) = \|\bar{G}_{\mathbf{a}}(z) - \bar{G}_{\mathbf{a}}(\tilde{z})\|_2. \quad (1)$$

The encoder distance is identical for every candidate action sequence; ACPC can additionally expose action-dependent amplification of that displacement. Whether this extra information matters is an empirical question, tested against one-step and action/time-destroyed controls rather than assumed from the definition.

Theorem 1 (Target-free prediction-error-drift bound). *Let $Y_{\mathbf{a}}^H$ be any projected external future generated from the same state and action sequence as both prediction branches. With*

$$e_h = \|\bar{G}_{\mathbf{a}}(E_\theta(h)) - Y_{\mathbf{a}}^H\|_2, \quad e_{\tilde{h}} = \|\bar{G}_{\mathbf{a}}(E_\theta(\tilde{h})) - Y_{\mathbf{a}}^H\|_2,$$

the following holds sample by sample:

$$|e_{\tilde{h}} - e_h| \leq \text{ACPC}_H(h, \tilde{h}, \mathbf{a}). \quad (2)$$

The adverse increase $(e_{\tilde{h}} - e_h)_+$ obeys the same bound, while the right-hand side requires no future target at diagnostic time.

Proof. This is the reverse triangle inequality in the weighted-stacked projected-rollout space. The common action sequence is required because both errors must share the same target. $\square$

3.2 Radius, discriminability guard, and calibration

For anchor i , let s_i be the median norm of its clean observed transitions, including the history-to-future boundary:

$$s_i = Q_{0.50}(\{\|u_{i,k}^{\text{obs}} - u_{i,k-1}^{\text{obs}}\|_2\}_{k=1}^H). \quad (3)$$

For probe draw m , the normalized radius and checkpoint statistic are

$$R^{(i,m)} = \frac{\|\bar{G}_{\mathbf{a}_i}(E_\theta(h_i)) - \bar{G}_{\mathbf{a}_i}(E_\theta(\tilde{h}_i^{(m)}))\|_2}{s_i + \varepsilon}, \quad \text{ATR}_q = Q_q\left(\left\{\frac{1}{M} \sum_{m=1}^M R^{(i,m)}\right\}_{i=1}^n\right). \quad (4)$$

We use $H = 8$, uniform horizon weights, and $q = 0.90$. ATR is an empirical upper-tail summary, not a candidate-distribution probability.

Low radius alone accepts a constant model. Selective Margin Pass Rate (SMPR) therefore measures how often programmatically selected different-state proxy pairs remain outside the same-state perturbation tube by a fixed normalized margin. The guard rejects complete collapse, but its controls do not establish semantic or action faithfulness; closed-loop behavior remains the authority.

For model family $\mathcal{F}$ and stressor v , calibration thresholds t_R, t_M define the higher-is-better score

$$S_{\mathcal{F},v}(\theta) = \min\left\{\frac{t_R - \text{ATR}_v(\theta)}{|t_R| + \varepsilon}, \frac{\text{SMPR}_v(\theta) - t_M}{|t_M| + \varepsilon}\right\}. \quad (5)$$

The endpoint rule $S \geq 0$ uses only one checkpoint at scoring time, but its thresholds are family-calibrated. Given a reference checkpoint θ_0 , the paired rule

$$\Delta S_{\mathcal{F},v}(\theta_1; \theta_0) = S_{\mathcal{F},v}(\theta_1) - S_{\mathcal{F},v}(\theta_0) > 0 \quad (6)$$

asks whether the endpoint improves under the same scoring map. Its zero threshold survives a common monotone rescaling within a pair, but the rule requires the reference and does not imply that either checkpoint is absolutely robust.

Definition 1 (Family-calibrated diagnostic region). *The diagnostic region is the set of checkpoints satisfying $\text{ATR} \leq t_R$ and $\text{SMPR} \geq t_M$ under one frozen family/stressor calibration. Its meaning is established empirically on held-out training seeds; raw thresholds are not assumed to transfer across model architectures.*

Appendix E gives a conditional candidate-cost interpretation for already evaluated ordered pools; it does not cover adaptive sampling or replanning. Appendix F relates ACPC to local encoder–predictor sensitivity.

4 Experimental Setup

Models, tasks, and stressors. LeWM [13] is an end-to-end JEPA world model used for latent-space model-predictive control. TwoRoom, PushT, Reacher, and Cube span navigation, planar manipulation, arm control, and 3D manipulation. We train no-noise checkpoints and full-sequence Gaussian-augmentation checkpoints with $\sigma_{\max} \in \{0.01, \dots, 0.08\}$. The primary endpoint adds observation noise $\sigma = 0.08$ while keeping the goal image clean. Off-axis tests use fixed Gaussian blur ($k = 15$) and resize (scale 0.25).

Behavior and diagnostic units. For each task and LeWM training seed 3072/3073/3074, the nine-point Gaussian grid is evaluated with seeds 42/43/44 and 100 trajectories per evaluation seed. Evaluation seeds are conditional measurement replicates; training seed is the model-replication unit. ATR uses 100 anchors, five probe draws, $H = 8$, uniform horizon weights, within-anchor draw averaging, and the checkpoint q90. SMPR uses the q90 tube, closest-35% state-proxy neighborhood, and margin 0.10. Protocol details are in Appendix B.

Future-drift prediction. The mechanism test uses 16 trajectory blocks per task, four probe severities, and two probe draws. Leave-one-block-out ridge models receive common severity, encoder, $H = 1$, action-magnitude, and nominal-dynamics covariates; $H = 8$ variants add the correct response or an action-zeroed, candidate-shuffled, or time-shuffled response. The held-out target is the corresponding observed future embedded by the frozen target encoder. Some seed 3073 task results preceded the final protocol freeze; seed 3074 is the fully frozen four-task replication. Reported cluster intervals are conditional on these two trained checkpoints.

Table 1: Held-out future-drift prediction on fragile no-noise LeWM checkpoints. Entries are positive MAE reductions (%) after adding correct-action H8 to common encoder+H1 covariates; destroyed H8 is the best action/time-destroyed variant. Seed 3074 is the fully frozen replication. Intervals resample task $\times$ trajectory blocks conditional on both trained checkpoints.

Task / aggregation	Absolute drift		Adverse drift	
	vs H1	vs destroyed H8	vs H1	vs destroyed H8
TwoRoom	57.5	52.6	57.3	52.4
PushT	29.0	28.3	25.1	23.6
Reacher	81.8	73.6	81.8	73.6
Cube	57.5	53.0	57.5	53.0
Seed 3073, equal-task mean	60.8	54.8	60.4	54.4
Seed 3074, equal-task mean	51.4	47.7	49.6	45.7
Two-seed mean [95% CI]	56.5 [51.1,61.2]	51.9 [45.0,56.6]	55.5 [50.1,60.2]	50.6 [44.0,55.3]

Frozen calibration and stressor transfer. The diagnostic region is fitted only on the 36 seed 3072 rows and applied unchanged to all 72 seed 3073/3074 rows. The stressor-transfer test scores every fixed-severity blur/resize base-endpoint pair before joining behavior: 24 pairs across four tasks, three training seeds, and two stressors. Positive transfer requires at least a five-point stressed-success gain and at most a five-point clean-success loss. The endpoint screen and paired $\Delta S > 0$ rule receive no stressor-specific retuning. Bootstrap resampling uses task $\times$ training-seed blocks and retains the two stressors together within each block.

5 Experiments

Gaussian evaluation noise first creates a controlled fragile/recovered family. We then test action-matched future drift, calibration across training seeds, and transfer to blur/resize in that order. Closed-loop success supplies behavioral labels but is never an input to the diagnostic score.

5.1 Behavioral recovery and diagnostic co-movement

At evaluation noise $\sigma = 0.08$, no-noise LeWM checkpoints lose 25.89 ± 2.38 , 74.56 ± 5.55 , 41.00 ± 1.44 , and 22.11 ± 2.78 success points on TwoRoom, PushT, Reacher, and Cube, respectively. Full-sequence Gaussian augmentation recovers broad, task-dependent regions rather than one universal best training strength. Figure 1 shows the observation-noise score together with the two frozen diagnostic components over all nine training levels and three training seeds.

ATR contracts and SMPR rises through the recovery region on every task. Cube is the useful boundary case: its diagnostics improve strongly while behavioral recovery is weaker than on PushT or Reacher. The panel therefore motivates a calibrated two-component screen, but behavior remains the outcome authority.

5.2 Action-matched rollouts predict future drift

Theorem 1 guarantees a bound for the correct action-matched pair. We additionally ask whether the correct $H = 8$ response adds held-out information after every comparator receives the same severity, encoder, $H = 1$, action-magnitude, and nominal-dynamics covariates. The controls replace $H = 8$ with responses whose action or temporal structure is destroyed.

Across the behaviorally fragile no-noise checkpoints, the two-seed equal-task reduction in absolute-drift MAE is 56.5% versus $H = 1$ and 51.9% versus the strongest structure-destroyed $H = 8$ control. Adverse drift gives reductions of 55.5% and 50.6%. Correct $H = 8$ beats $H = 1$ and every destroyed control in 57/64 seed-averaged task $\times$ trajectory clusters; every task passes separately in both seeds. The fully frozen seed 3074 replication passes all four tasks for both targets.

These intervals resample task $\times$ trajectory blocks while retaining the two trained checkpoints; they quantify conditional block uncertainty, not population uncertainty over training seeds. The result establishes action/horizon increment for logged futures in fragile regimes. Appendix C reports range and candidate-query controls that delimit this interpretation.

Table 2: Held-out transfer of a LeWM calibration fit on seed 3072 and applied without refitting to independently trained seeds 3073/3074. BA denotes balanced accuracy; AUPRC denotes area under the precision–recall curve.

Evaluation set	rows	BA	AUPRC	precision	recall
Both test seeds	72	0.945	0.965	0.961	0.980
Seed 3073	36	0.931	0.993	0.962	0.962
Seed 3074	36	0.958	0.941	0.960	1.000

Table 3: Transfer to all 24 fixed-severity LeWM blur/resize pairs without stressor-specific retuning. BA is balanced accuracy and AUPRC is area under the precision–recall curve. The single-checkpoint region accepts 7/24 endpoints (selective false-pass-risk U95: 0.348); paired change requires the no-noise reference.

Rule	BA	AUPRC	precision	recall	ρ_s
Single-checkpoint region	0.733	0.856	1.000	0.467	–
Paired change ($\Delta S > 0$)	0.889	0.974	0.882	1.000	0.864

5.3 Calibration transfers across training seeds

Thresholds fitted only on the 36-row seed 3072 Gaussian calibration grid are applied unchanged to the corresponding 72 rows from independently trained seeds 3073/3074. The diagnostic region reaches balanced accuracy/AUPRC 0.945/0.965, precision 0.961, and recall 0.980.

A 5,000-repeat bootstrap over the eight task×training-seed test blocks gives intervals [0.893, 0.991] for balanced accuracy and [0.904, 1.000] for AUPRC. Deleting any task without refitting leaves balanced accuracy in [0.924, 0.956], and the two test seeds reach 0.931 and 0.958. This is a same-task-family cross-seed replication: it does not establish unseen-task, cross-family, or universal numerical calibration.

5.4 Gaussian calibration transfers to blur and resize

The fixed-strength evaluation includes every LeWM base–endpoint pair under blur and resize: four tasks, three independently trained seeds, and two stressors. Neither the Gaussian-calibrated endpoint screen nor the paired rule $\Delta S > 0$ is stressor-retuned.

The shared paired rule reaches balanced accuracy 0.889, AUPRC 0.974, precision 0.882, recall 1.000, and Spearman correlation 0.864. Block bootstrap intervals are [0.700, 1.000] for balanced accuracy, [0.869, 1.000] for AUPRC, and [0.641, 0.932] for Spearman correlation. Without retuning, balanced accuracy is 0.875 on blur and 0.900 on resize. A block sign-flip test gives one-sided $p = 2.44 \times 10^{-4}$.

The endpoint screen uses no base diagnostic at scoring time, but its evaluation label remains pair-derived improvement. It passes 7/24 endpoints (coverage 0.292), with no observed false pass, recall 0.467, and an exact one-sided 95% false-pass upper bound of 0.348. It is therefore a conservative sample screen, not an absolute stability detector or population safety guarantee. Component comparisons in Appendix G show that this checkpoint-change association is not uniquely action specific; the action/horizon increment is established only by the future-drift experiment.

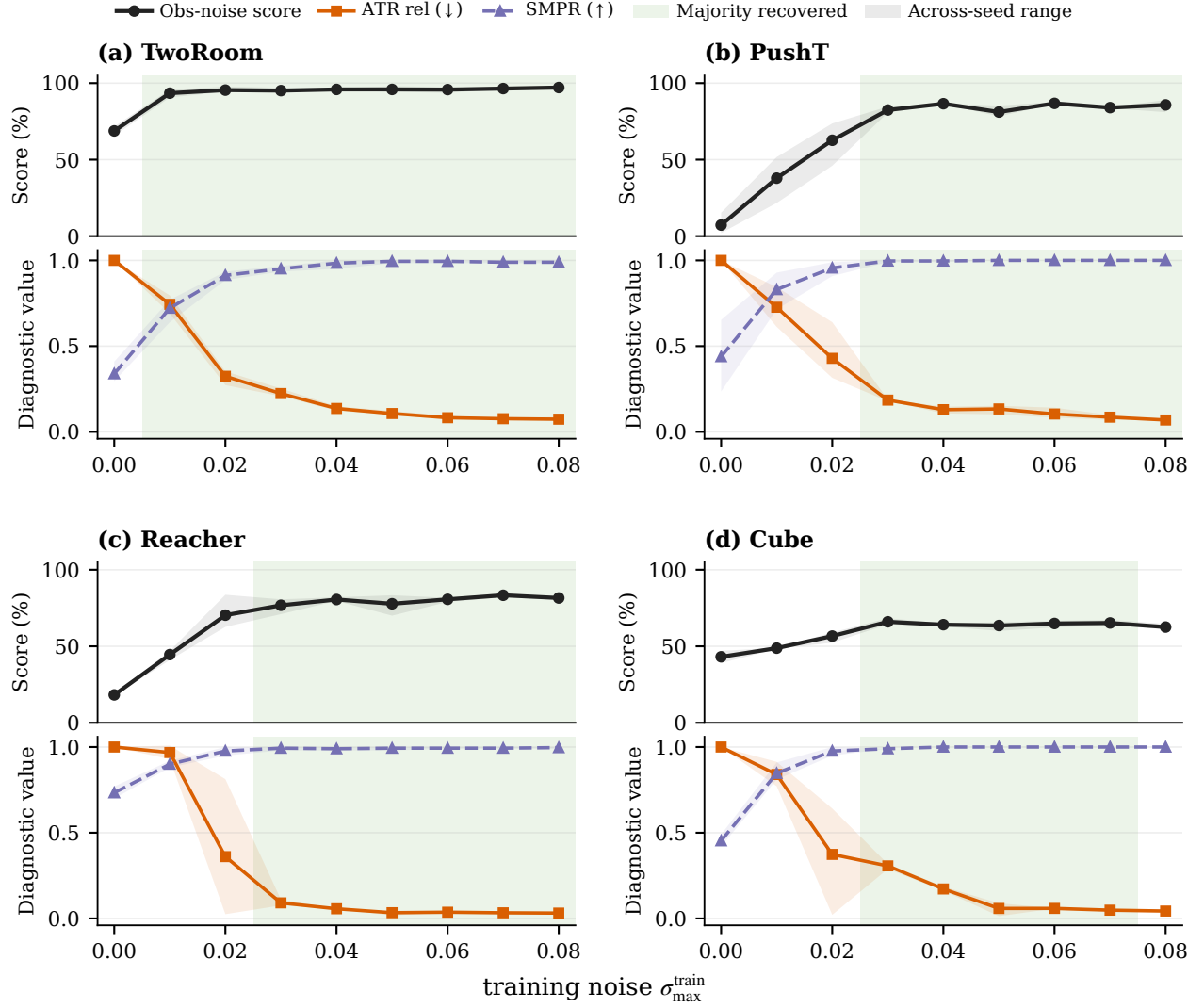


Figure 1: Closed-loop success under observation noise and the two diagnostic components across the Gaussian training sweep. Solid lines first average evaluation replicates within each training seed and then average seeds 3072/3073/3074; translucent envelopes span the three seed means. Green vertical bands mark training levels where a majority of seed rows meet the recovery criterion. Lower ATR and higher SMPR co-move with recovery on all four tasks, but this descriptive alignment alone does not establish action or semantic specificity.

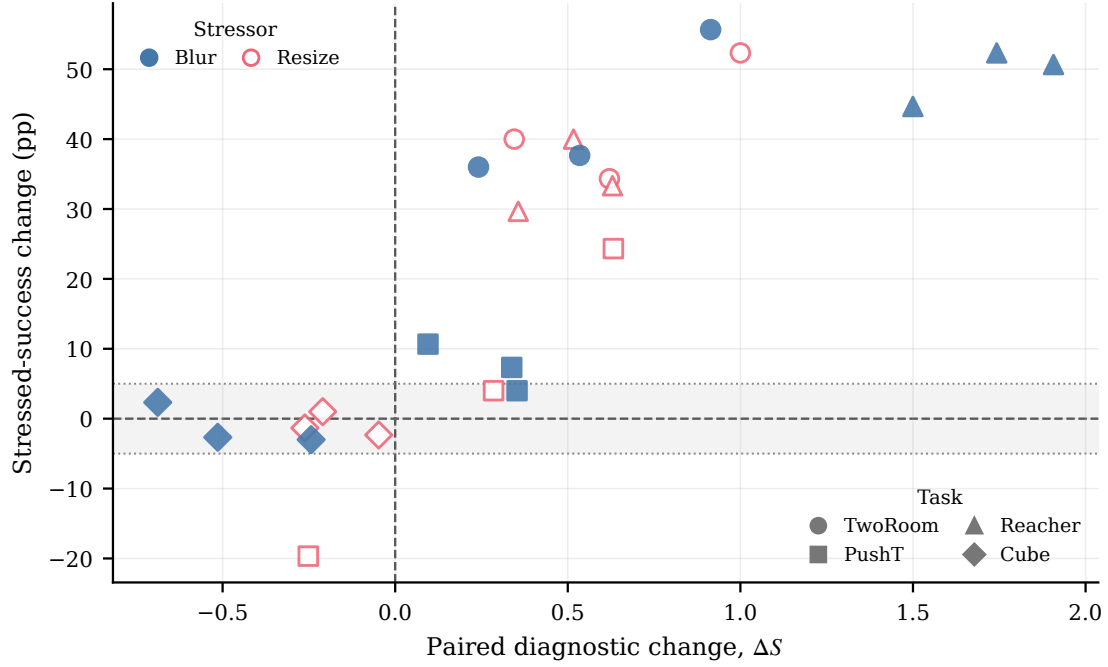


Figure 2: LeWM endpoint-minus-base diagnostic change versus stressed-success change for all 24 fixed blur/resize pairs. Color and fill identify stressor; marker shape identifies task. Dashed lines mark zero change, and the shaded strip marks the predeclared ± 5 -point behavioral neutral band. Positive-transfer labels additionally require a clean-success drop no larger than five points, which is not shown on this axis.

6 Discussion and limitations

The future-drift experiment validates the object in Theorem 1: correct-action $H = 8$ contains information about action-matched latent future-error drift that $H = 1$ and structure-destroyed controls miss in fragile regimes. Its target is a held-out future embedded by the frozen encoder, not behavior. The interval is conditional on two trained checkpoints: some seed 3073 task results preceded the final protocol freeze, whereas seed 3074 is a fully frozen four-task replication. Candidate-query increment remains task dependent.

Cross-seed calibration changes the independently trained model instance while holding the four tasks and training-noise grid fixed. It therefore supports family-local replication, not unseen-task generalization, cross-family numerical transfer, a deterministic single-frame verdict, or a universal latent threshold.

Cross-stressor transfer concerns relative checkpoint improvement at one fixed blur and resize severity. The endpoint screen uses no base diagnostic at scoring time, yet its label is pair-derived and its 7/24 coverage leaves a wide false-pass upper bound. The stronger paired result requires a reference checkpoint and is not action specific. Neither result is an absolute cross-stressor stability detector.

Across all three experiments, SMPR is a gross-collapse guard rather than a semantic proof, the fixed-pool analysis applies only after both candidate pools are evaluated, and stable-but-wrong predictions remain possible. Closed-loop evaluation is therefore the behavioral authority. The highest-value extensions are additional prospectively frozen training seeds for the future-drift test, absolute cross-stressor labels over multiple severities, and independently trained external-model families.

7 Conclusion

ACPC converts a paired visual probe into a target-free bound on action-matched future-error drift. On four tasks, its action-conditioned long-horizon response adds held-out mechanism information in fragile checkpoints; a one-seed LeWM calibration replicates across two independent training seeds; and one paired rule retains useful ordering under fixed blur and resize. Keeping these claims separate yields a compact diagnostic with explicit statistical and operational boundaries rather than a universal robustness score.

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A Candidate-Cost and Fixed-Pool Analysis

These details connect the rollout radius to candidate-cost drift and fixed-pool top-1 stability; they do not provide closed-loop control guarantees. Let h and $\tilde{h}$ be matched clean and corrupted histories, and let both branches evaluate the same ordered candidate pool $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$. For task descriptor g , define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \dots, \Pi(\hat{z}_{t+H}), g), \quad (7)$$

with $C_{\tilde{h}}$ defined analogously. If j^* is the clean winner, let $\Delta_j = C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{j^*}, g)$ and $d_j = |C_{\tilde{h}}(\mathbf{a}^j, g) - C_h(\mathbf{a}^j, g)|$. The candidate-wise sufficient condition used below is

$$\min_{j \neq j^*} (\Delta_j - d_j - d_{j^*}) > 0. \quad (8)$$

Proposition 1 (ACPC controls candidate-cost drift). *Assume that $J(\cdot, g)$ is locally L_J -Lipschitz on the paired projected-rollout neighborhood evaluated by the fixed candidate set. If*

$$\|\bar{G}_{\mathbf{a}}(E_\theta(h)) - \bar{G}_{\mathbf{a}}(E_\theta(\tilde{h}))\|_2 \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

Proof. Apply the Lipschitz condition to the two arguments of J and use the assumed bound. $\square$

Proposition 2 (Candidate top-1 stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j ,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. $\square$

Candidate-wise sharp condition. For clean winner j^* and any competitor j , the drift definition gives

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{j^*}, g) \geq \Delta_j - d_j - d_{j^*}.$$

Therefore positivity of the minimum in Equation (8) makes every perturbed competitor strictly more costly. This proves the deterministic fixed-pool implication without a common worst-case drift bound.

Local Gaussian sensitivity. For small Gaussian pixel noise ξ , a first-order expansion gives $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Evaluation and Diagnostic Protocol

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels

with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \dots, 0.08\}$.

ATR is computed at each task-training-seed-checkpoint row from one canonical weighted-stacked horizon radius per anchor. The default is $H = 8$ with uniform $\alpha_k = 1/H$. Each radius is divided by that anchor’s q50 clean observed-transition scale, including the transition from the last history observation to the first future observation; multiple noise draws are first averaged within the anchor, and q90 is then taken across anchors. SMPR uses programmatic label-crossing state pairs, q90 tube radius, closest-35% neighborhood, and normalized margin 0.10. The paper reports population mean/std across training seeds 3072/3073/3074.

All summary figures and tables are rebuilt deterministically by the scripts documented in `paper1/scripts/REA DME.md`. Checkpoint-level diagnostics additionally require the released checkpoints and datasets. Machine-readable summaries record training seeds, evaluation seeds, and protocol hashes.

C Future-Drift Scope and Controls

The target-free inequality in Theorem 1 has zero numerical violations in every logged-H8 and candidate-H5 task×seed×checkpoint-role cell. This is an implementation invariant, not an independent statistical success count.

Endpoint-only comparisons are range restricted, and earlier coarse checkpoint-label screens leave simple encoder or one-step diagnostics competitive. Those targets do not test the future-drift question: incremental prediction of an independent action-matched future after common covariates are supplied. On the candidate- $H = 5$ query, the increment is also task dependent: seed 3073 clears it on TwoRoom and PushT but not Reacher or Cube; seed 3074 clears the absolute target on TwoRoom and PushT and the adverse target only on PushT. We therefore make no four-task candidate-query increment claim.

These controls restrict the main claim to logged future-error drift in fragile regimes. The primary estimate keeps seeds 3073/3074: some seed 3073 task runs preceded the final protocol freeze, whereas seed 3074 is the fully frozen replication. After outcomes had been inspected, seed 3072 was run as a retrospective completeness check; it passes logged $H = 8$ on all four tasks but is reported separately and is not pooled with the primary seeds.

D Qualitative Neighborhood Visualization

The two-dimensional view complements the quantitative analysis by showing how repeated perturbations of the same anchors are arranged before and after noise training. Because t-SNE does not preserve metric geometry, all reported evidence remains in the original projected-rollout space.

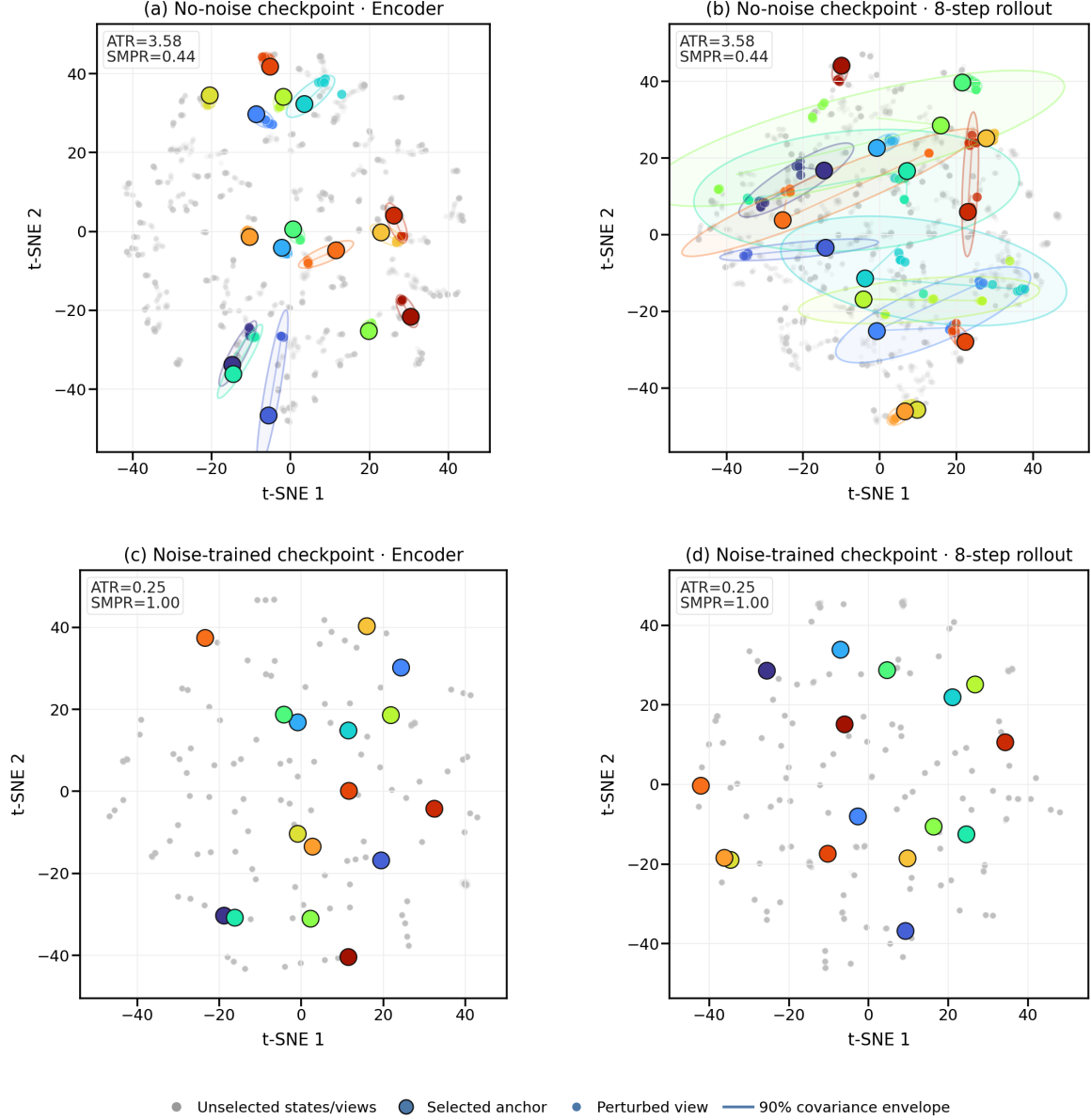


Figure 3: Qualitative PushT neighborhoods under matched visual perturbations. The panels embed the same fixed anchors and repeated views for the no-noise and $\sigma_{\max} = 0.08$ checkpoints, in encoder and eight-step rollout spaces. Colored neighborhoods make the contraction after noise training visually accessible. The embedding protocol is fixed across panels, but t-SNE is not metric preserving: no displayed distance, area, or overlap enters a claim, and ATR/SMPR are computed in the original projected-rollout space.

E Fixed-Pool Candidate Stability

The analysis evaluates the same ordered pool of $K = 65$ actions on nominal and corrupted branches for 100 histories at each of 108 LeWM checkpoints (10,800 histories). Coarse coverage is 0.478, candidate-wise sharp coverage is 0.700, and the overall observed top-1 flip rate is 0.208. Among sharp-condition failures the flip rate is 0.694.

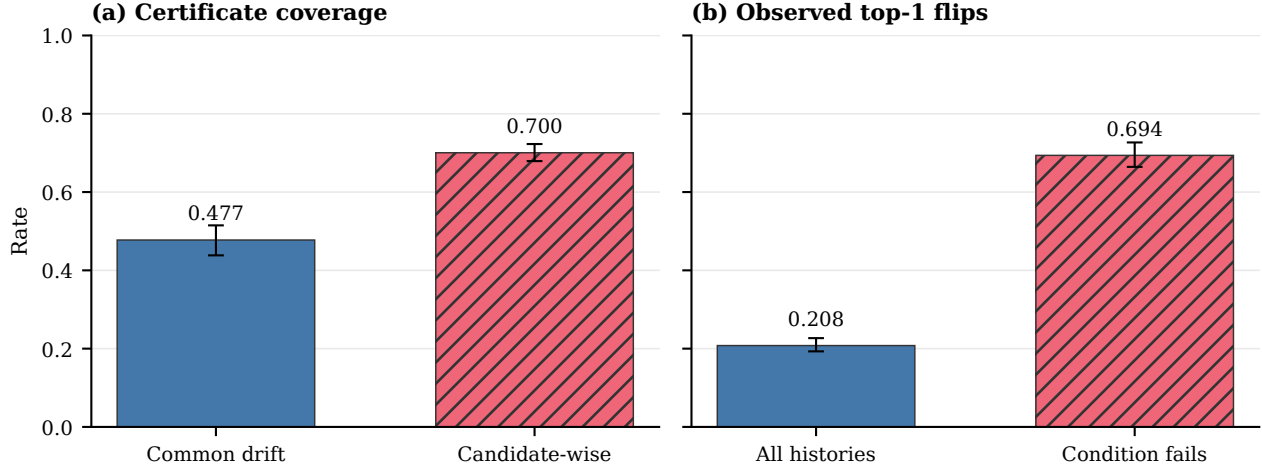


Figure 4: Fixed-pool candidate-stability conditions with task $\times$ training-seed block-bootstrap 95% intervals. Left: coverage of the common-drift and candidate-wise sharp conditions. Right: observed top-1 flip rate over all histories and over histories that fail the sharp condition. The zero flip rate when the sharp condition holds follows deterministically and is not counted as separate validation evidence.

This analysis is post-evaluation: candidate-specific drift requires both branches and is not a cheap predictor for a new pool. Extending it to adaptive CEM would require prospective calibration over newly sampled pools and repeated replanning.

F Local Linearization

For small raw-pixel Gaussian noise ξ , the paired rollout difference has first-order term $J_{\bar{G}_a} J_E \xi$. The covariance-aware JVP/Hutchinson estimate uses the same weighted $H = 8$ map, anchors, and ImageNet-coordinate correction as the finite-difference measurement.

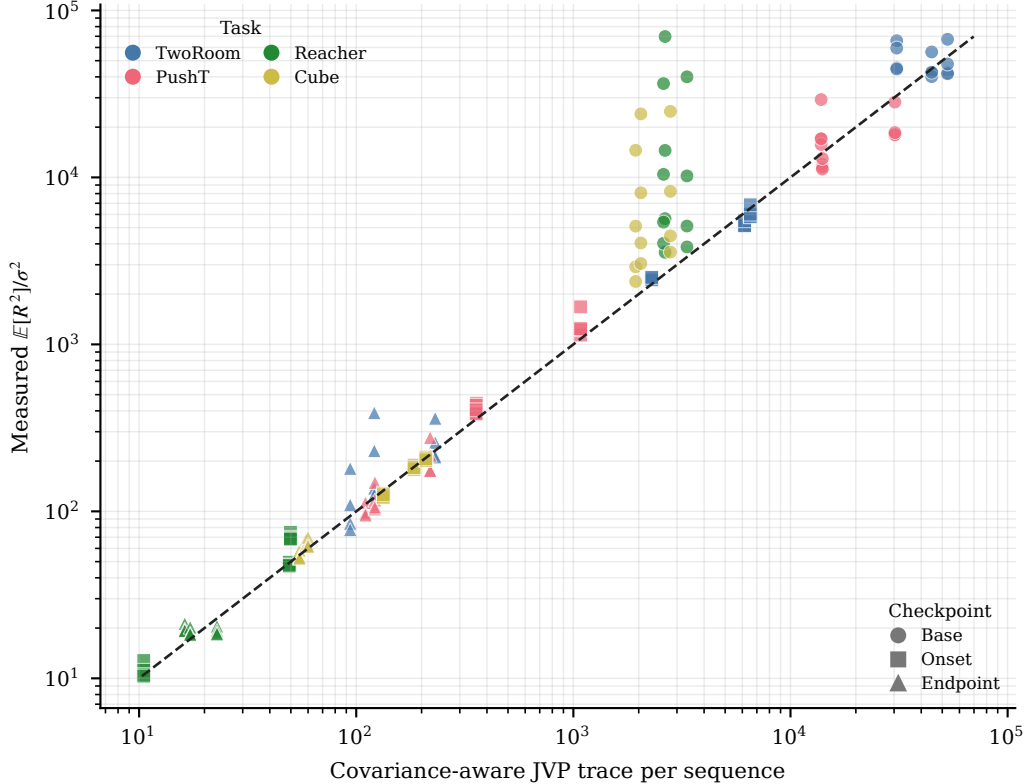


Figure 5: Finite-difference $\mathbb{E}[R^2]/\sigma^2$ versus the covariance-aware composed JVP trace for 36 checkpoints and four small noise levels. The dashed line is equality; deviations for several no-noise Reacher/Cube checkpoints show that the finite tested range is not uniformly linear.

The endpoint composed trace contracts strongly, but rollout-side traces and finite-range linearization error are task dependent. These diagnostics explain a local mechanism; they do not provide a global or closed-loop certificate.

G Cross-Stressor Component Analysis

The analysis includes all 24 fixed-severity LeWM blur/resize base–endpoint pairs. Component scores are competitive: Spearman correlations with robustness change are 0.887 for encoder change, 0.868 for correct-action H8, 0.875 for time-shuffled H8, and 0.864 for the joint score. Thus the cross-stressor result supports predictive ordering within LeWM, not a uniquely action-specific mechanism; action/horizon increment is claimed only for future-drift prediction.

Table 4: Cross-stressor component comparison on all 24 LeWM blur/resize pairs. Choice accuracy selects the endpoint iff the oriented diagnostic change is positive; regret is stressed-success loss against the better of base and endpoint. The exact 12-block sign-flip test gives one-sided $p = 0.0002$. Remaining-set Spearman ranges after deleting one task and one training seed are $[0.748, 0.897]$ and $[0.753, 0.940]$. Competitive component rows show that paired ordering transfers without establishing a unique action-specific mechanism.

Diagnostic	BA	AUPRC	ρ_s	choice acc.	regret (pp)
Encoder q90	0.778	0.992	0.887	0.917	0.917
H1 q90	0.778	0.957	0.763	0.917	0.917
Correct-action H8 q90	0.889	0.988	0.868	0.917	0.139
Action-shuffled H8 q90	0.833	0.981	0.832	0.875	0.958
Action-zeroed H8 q90	0.833	0.988	0.870	0.875	0.958
Time-shuffled H8 q90	0.889	0.992	0.875	0.917	0.139
SMPR	0.911	0.992	0.913	0.833	0.750
Joint score	0.889	0.974	0.864	0.917	0.139